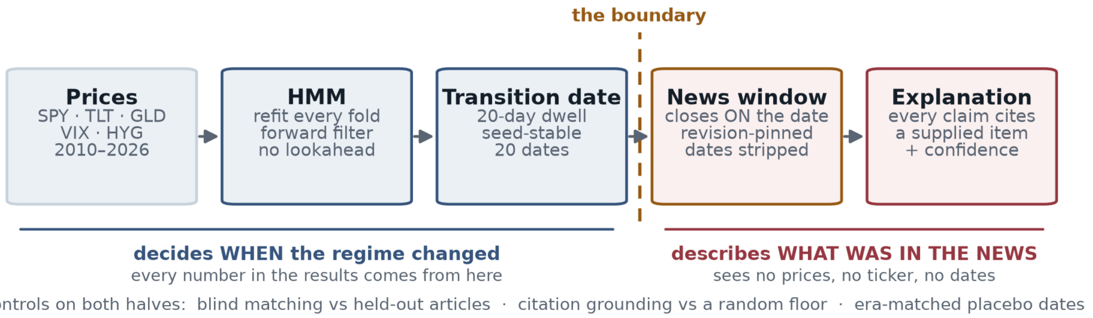
Regime detection that can say what happened — and a measurement of how much that is worth

An auditable explanation layer for any regime model. BSc Mathematics with Statistics, University of Bristol.

The claim. A two-state HMM refitted inside every walk-forward fold separates SPY volatility regimes in **18 of 20 folds** containing both states (sign test p = 0.00040). Reading only news published *before* each of 20 transitions, a language model produced cited explanations matching back to their own 14-day window **72% of the time against 33% chance**, with **96.4% of 474 claims grounded, zero fabricated**. Whether they are *diagnostic* is unresolved: **+18.5pp, 95% CI [-12, 47]pp**.



PROBLEM STATEMENT

A regime model tells you the market changed, not what happened — it only ever sees returns. So its output is coloured bands that a human interprets from memory. On a desk of thirty portfolio managers that is thirty private, unrecorded readings of one signal, and the reasoning leaves with whoever remembers the week.

SOLUTION OVERVIEW

Above. A two-state HMM marks *when* the regime changed; a language model reading *only news published before that date* describes what was happening. The two never mix: no statistic comes from the model, and it never moves a regime boundary.

The boundary is enforced by the type system — a window object refuses to construct if any item post-dates it. Pages are pinned to their revision id, because a mean of **38.2%** of a Wikipedia day-page as it stands today (peak 71.2%) was written *after* the day it describes. Fetching the live page feeds the model hindsight, invisibly.

USE OF AI

`claude-opus-5` via the Anthropic Messages API — one call per window, effort high, output schema enforced server-side. Prompts live in version-controlled files with a dated iteration history, never inline strings; every call logs the model id, prompt hash and input hash, so any sentence in the results traces to the exact prompt and news window behind it. Built with Claude Code. Wikipedia MediaWiki API, yfinance, hmmlearn, scikit-learn, Streamlit.

IMPACT & VALUE

It replaces "trust me, that band is the Greek referendum" with a written explanation citing dated sources — **arriving with its own error bars.** Tested on 20 transitions and 40 matched control dates:

Does the narrator work?	Result	Against
Matched to its own 14-day window	72%	33% chance
Claims grounded in the item cited	96.4%	1.8% floor
Fabricated citations	0	of 474
Confident on real transitions	**80%**	**vs 62% ordinary days**
Same tests, a different AI	same	scores repeat

The reusable asset is the controls, not the explanations. Anyone can ask a model what happened in a month; nobody can otherwise say whether the answer is specific to it. Point this at any regime model's dates and every explanation returns with that table attached.

REFLECTIONS

Row four is the finding. The model explains an ordinary Tuesday almost as readily as a real transition. Calling that "not diagnostic" would be a no-difference claim $p = 0.43$ does not license: it is **+18.5pp, 95% CI [-12, 47]pp**, unresolved — power at that gap is only 0.17, and the constraint is 20 transitions, not controls, so more placebos could never have helped. **What this buys today is auditability, not signal generation.**

Next: pre-specify volatility *onsets* rather than all transitions, which roughly triples the power. One leak stays open — the boundary is 23:59 UTC and Wikipedia reports the US close the same evening, though controls carry *more* of it than transitions.